

Page-8 FIRST PRE-BOARD EXAM. (2024-25)**CLASS X
(Science 086)****Max. Marks: 80****Time: 3 hours**

General Instructions:

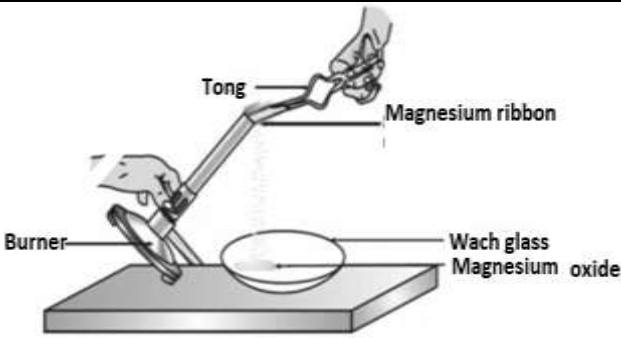
Read the following instructions carefully and strictly follow them.

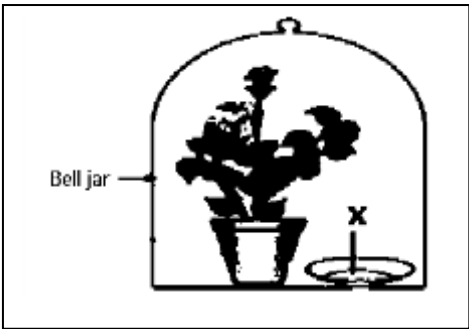
- (i) This question paper consists of 39 questions. All questions are compulsory.
- (ii) Question paper is divided into five sections A, B, C, D and E.
- (iii) In Section A -question number 1 to 20 are Multiple Choice Questions (MCQs) carrying 1 mark each.
- (iv) In Section B - question number 21 to 26 are Very Short Answer (VSA) type questions carrying 2 marks each. Answer to these questions should be in the range of 30 to 50 words.
- (v) In Section C - question number 27 to 33 are Short Answer (SA) type questions carrying 3 marks each. Answer to these questions should be in the range of 50 to 80 words.
- (vi) In Section D - question number 34 to 36 are Long Answer (LA) type questions carrying 5 marks each. Answer to these questions should be in the range of 80 to 120 words
- (vii) In Section E - question number 37 to 39 are of 3 source-based/case-based units of assessment carrying 4 marks each with sub-parts.
- (viii) There is no overall choice. However, an internal choice has been given in some sections.

SECTION A

Select and write one most appropriate option out of the four options given for each of the questions 1-20

1	Junction between two neurones is called (a) cell junction (b) neuromuscular junction (c) neural joint (d) synapse	1
2	At what temperature is gypsum heated to form Plaster of Paris? (A) 35°C (B) 70°C (C) 80°C (D) 100°C	1

3	Which of the following is an antacid a. HCl b. H_2SO_4 c. NaCl d. $\text{Mg}(\text{OH})_2$	1
4	 Which of the following is the correct observation of the reaction shown in the above set up? (a) Brown powder of Magnesium oxide is formed. (b) Colourless gas which turns lime water milky is evolved. (c) Magnesium ribbon burns with brilliant white light. (d) Reddish brown gas with a smell of burning Sulphur has evolved	1
5	In the binary fission method of multiplication : (a) only one parent is involved (b) no gametes are formed (c) fertilisation does not take place (d) all the above statements are true	1
6	Which is the first enzyme to mix with food in the digestive tract? (a) Pepsin (b) Cellulose (c) Amylase (d) Trypsin	1
7	Choose the incorrect statement about insulin (a) It is produced from pancreas (b) It regulates growth and development of the body. (c) It regulates blood sugar level (d) Insufficient secretion of insulin will cause diabetes	1
8.	Principal focus of a spherical mirror lies (a) midway between the pole and centre of curvature (b) near the pole (c) near the centre of curvature (d) None of these	1

9	Where should an object be placed in front of a convex lens to get a real image of the same size of the object ? (a) At the principal focus of the lens (b) At twice the focal length (c) At infinity (d) Between the optical centre of the lens and its principal focus.	1
10	What is the current through a 5.0 ohm resistor if the voltage across it is 10 V (a) zero (b) 0.5 A (c) 2.0 A (d) 5.0 A	1
11	The contraction and expansion movement of the walls of the food pipe is called: (a) Translocation (b) Transpiration (c) Peristaltic movement (d) Digestion	1
12	Observe the experimental set up shown below. Name the chemical indicated as 'X' that can absorb the gas which is evolved as a by product of respiration. 	1
13	Which of the following option shows the transport of oxygen to the cell correctly? (a) Lungs→pulmonary vein→left atrium →left ventricle →aorta → body cells (b) Lungs→pulmonary vein→right atrium→right ventricle→aorta→body cells (c) Lungs →pulmonary artery →left atrium → left ventricle → vena cava → body cells	1

	(d) Lungs→pulmonary artery→right atrium→right ventricle→vena cava→body cells	
14	The strength of magnetic field inside a long current carrying straight solenoid is (a) more at the ends than at the centre. (b) minimum in the middle (c) same at all points (d) found to increase from one end to the other	1
15	Identify the incorrect statement 'The energy available to the producers is maximum' because : (a) It is the first trophic level which absorbs 1% of light energy directly from the source. (b) It utilizes the most of the chemical energy for its own respiration, growth, reproduction, movement etc. (c) It utilizes 10% of light energy and transfers the rest to the next trophic level. (d). It transfers only 10% of light energy to the next trophic level.	1
16	Vegetative propagation refers to formation of new plants from (a) stem, roots and flowers (b) stem, roots and leaves (c) stem, flowers and fruits (d) stem, leaves and flowers	1

Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(A). If both A and R are true, and R is the correct explanation of A.

(B). Both A and R are true, and R is not the correct explanation of A.

(C). A is true but R is false.

(D). A is false but R is true

17	Assertion(A) : The sex of the children will be determined by chromosome received from the father. Reason : A human male has one X and one Y –chromosome which play important role in sex determination.	1
18	Assertion(A): Carbon and its compound are used as fuels for most applications. Reason(R):Carbon and its compounds can easily burn in air at a moderate rate, produce large amount of heat energy and pollute the atmosphere marginally.	1
19	Assertion (A): A convex mirror always forms an image behind it and the image formed is virtual. Reason (R): According to the sign convention, the focal length of a convex mirror is positive.	1
20	Assertion (A): If the lions are removed from a food chain it will not affect the	1

	food chain, however if the plants are removed from a food chain it will disturb the ecosystem. Reason (R): Plants are producers who can make food using sunlight, while lions are consumers.	
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SECTION-B

Question no. 21 to 26 are very short answer questions.

21	i) $\textcircled{A} + \textcircled{BC} \rightarrow \textcircled{AC} + \textcircled{B}$ ii) $\textcircled{AB} + \textcircled{CD} \rightarrow \textcircled{AC} + \textcircled{BD}$ Identify the types of reaction mentioned above in(i)and(ii).	2
22	Name the final product/products obtained in the anaerobic respiration, if it takes place: (a) In a plant (like yeast). (b) In an animal tissue (like muscles). OR The leaves of a plant first prepare food A by photosynthesis. Food A then gets converted into food B. What are A and B?	2
23	Show the formation of NaCl and MgO by transfer of electrons. OR Give reason: (i) Ionic compounds have high melting point. (ii) Ionic compounds are hard crystalline solids.	2
24	When metal X is treated with a dilute acid Y, then a gas Z is evolved which burns readily by making a little explosion. (a) Name any one metals which can behave like metal X. (b) Name any one acids which can behave like acid Y. (c) Name the gas Z. (d) Is the gas Z lighter than or heavier than air?	2
25	A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging?	2
26	Why did Mendel choose pea plant for his experiments. ?	2

SECTION C

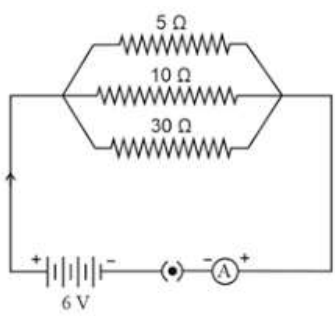
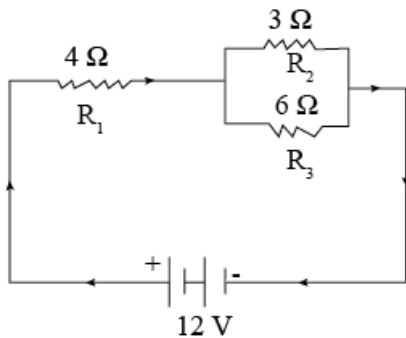
Question no. 27 to 33 are short answer questions.

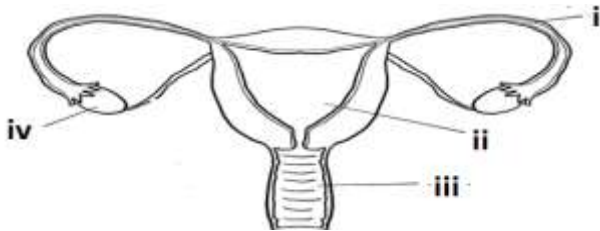
27	What are decomposers? List two important roles they play in the environment	3
28	What is a homologous series? Explain with an examples.	3
29	What is meant by scattering of light? Use this phenomenon to explain why the clear sky appears blue.	3

30	What is solenoid? Draw a sketch to show the magnetic field pattern produced by a current carrying solenoid? OR State one main difference between A.C. and D.C. Why is A.C. preferred over D.C. for long range transmission of electric power? Name one source each of D.C. and A.C.	3
31	How is bleaching powder prepared? Write the chemical equation of the reaction involved in the preparation of bleaching powder.	3
32	Explain Mendel's experiment with peas on inheritance of characters, considering only one visible contrasting characters.	3
33	An electric refrigerator rated 400W operates 8 hours per day. What is the cost of energy to operate it for 30 days at 3.00 per kWh?	3

SECTION D

Question no. 34 to 36 are Long answer questions.

34	In the circuit diagram given below, three resistors R_1, R_2, and R_3 of $5\ \Omega$, $10\ \Omega$, and $30\ \Omega$ respectively, which have been connected to a 12V battery. 3+1+1  Calculate: (i) the current through each resistor. (ii) total current in the circuit. (ii) the total circuit resistance. OR The circuit diagram given below shows the combination of three resistor R_1, R_2 and R_3. 2+2+1  Find (i) the total resistance of the circuit.	
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	(ii) the total current flowing in the circuit. (iii) the total potential difference across R1	
35	a) Label the parts indicated with i, ii, iii and iv : 2+3  b. How does reproduction help in providing stability to population of species? Or What are the different categories of contraceptive methods? Write briefly about each.	
36	Elaborate on the process of cleansing action. Illustrate micelle with the help of labelled diagram. (iii) Write the chemical equation of the preparation of soap from an ester $\text{CH}_3\text{COOCH}_3$. What is the name of this process? OR A salt X is formed and a gas is evolved when ethanoic acid reacts with sodium hydrogen carbonate. Name the salt X and the gas evolved. Write the balanced chemical equation of the reaction involved.	5

SECTION E

Question No. 37 to 39 are case-based/data -based questions.

37	On the basis of reactivity of different metals with oxygen, water and acids as well as displacement reactions, the metals have been arranged in the decreasing order of their reactivities. This arrangement is known as activity series or reactivity series of metals. The basis of reactivity is the tendency of metals to lose electrons. If a metal can lose electrons easily to form positive ions, it will react readily with other substances. Therefore, it will be a reactive metal. On the other hand, if a metal loses electrons less rapidly to form a positive ion, it will react slowly with other substances. Therefore, such a metal will be less reactive. (i) Which of the following metals is less reactive than hydrogen? (a) Copper (b) Zinc (c) Magnesium (d) Lead (ii) Which of the following metals is more reactive than hydrogen? (a) Mercury (b) silver (c) Iron (d) Gold (iii) Why is sodium kept immersed in kerosene oil? OR	4
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	What is the action of water on sodium and magnesium?	
38	Nastic movements in plants are not directional movements. They are not dependent on the stimulus and are growth independent. For example, the leaves of a touch me not plant fold up immediately when touched. These kinds of changes occur due to the changes in the amount of water in the leaves. Depending on the quantity, they either swell up or shrink. Plant hormones or phytohormones are responsible for the control and Coordination of plants. There are different types of hormones, which affect the growth of a plant. Phytohormones are chemical compounds which are released by stimulated cells. These hormones are diffused around the plant Cells. They have a role in the cell division, cell enlargement, cell differentiation, fruit growth, falling of leaves, ripening of fruits, ageing of plants etc. (i) Name the phenomenon called for the movement in growth of plants. (ii) What do you mean by nastic movement? (iii) What are the different types of hormones of plants? Or Write the function of gibberellin hormone.	4
39	While dealing with the reflection of light by spherical mirrors, we shall follow a set of sign conventions called the New Cartesian Sign Convention. In this convention, the pole (P) of the mirror is taken as the origin. The principal Axis of the mirror is take as the x -axis of the coordinate system. In a spherical mirror, the distance of the object from its pole is called the object distance (u). The distance of the image from the pole of the mirror is called the Image distance (v). Magnification produced by a spherical mirror gives the relative extent to which the image of an object is magnified with respect to the object size. It is expressed as the ratio of the height of the image to the height of the object. It is usually represented by the letter (m). (i) How can you calculate the magnification of a spherical mirror? (ii) What does a negative sign in the value of magnification indicates? (iii) Draw a ray diagram to show the formation of image of an object placed beyond the centre of curvature of a concave mirror. OR Why does the height of the object is taken to be positive?	4